



Exam Ace Champs

Where Champions Are Made

Quick Revision

Chapterwise

Mind-Maps

CHEMISTRY

**class
11**

Covers

all chapters of NCERT



Faster Recall



Quick Revision

Class : 11th Chemistry
Chapter- 1: Some Basic Concepts of Chemistry

How To Determine Empirical and Molecular Formula:

Step 1: Conversion of mass percent to grams.
 Step 2: Convert into number moles of each element.
 Step 3: Divide the mole value obtained above by the smallest number.
 Step 4: Write empirical formula by mentioning the number after writing the symbols of respective elements.
 Step 5: Writing molecular formula
 Determine empirical formula mass. Add the atomic masses of various atoms present in the empirical formula.
 Divide molar mass by empirical formula mass.

Empirical Formula: Simplest number ratio of various atoms present in a compound.
Molecular Formula: Shows exact number of different types of atoms present in a molecule of a compound.

Limiting Reagent: Reactant which gets consumed and limits the amount of product formed.

Mole Fraction

$$\text{Mole Fraction of A} = \frac{n_a}{n_a + n_b}$$

$$\text{Mole Fraction of B} = \frac{n_b}{n_a + n_b}$$

Homogenous: Uniform composition (Sugar solution, air)

Heterogenous: Non-uniform composition (Mixtures of salt and sugar.)

Elements: Contains one type of particles i.e., atoms, molecules (Na, Cu)

- (i). **Gay Lussac's Law of Gaseous Volume:** When gases combine or all produced in a chemical reaction they do so in simple ratio by volume provided all gases are at same temperature and pressure.
- (ii). **Avogadro Law:** Equal volumes of gases at the same temperature and pressure should contain equal number of molecules.
- (iii). **Dalton's Atomic Theory:**
 - Matter consists of indivisible atoms.
 - All the atoms of a given element have identical properties including identical mass.
 - Compounds are formed when atoms of different elements combine in a fixed ratio.
 - Atoms are neither created nor destroyed in chemical reaction.

How To Balance A Chemical Equation:

Step 1: Write correct formulas of reactants and products.
 Step 2: Balance number of C atoms.
 Step 3: Balance number of H atoms.
 Step 4: Balance number of O atoms.
 Step 5: Verify the number of atoms of each element.

Deals with the calculation of masses of the products and reactants involved in a reaction.

Stoichiometry

Atomic Mass Unit (amu): A mass exactly equal to one-twelfth the mass of one carbon-12 atom.

Molecular Mass: Sum of atomic masses of the elements present in a molecule.

Formula Mass: Sum of the atomic weights of atoms in a molecule. One Mole is the amount of a substance that contains as many particles/entities as there are atoms in

exactly 12g (or 0.012 kg) of the isotope. ¹²C

Molar Mass: Mass of one mole of a substance in grams.

Mass Percent

$$= \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

Molarity

$$= \frac{\text{No. of moles of solute}}{\text{Mass of solute in kg}}$$

$$= \frac{\text{No. of moles of solute}}{\text{Volume of solution in litres}}$$

Some Basic Concepts of Chemistry

Importance of Chemistry

In meeting human needs for food, health care products and other products required for improving quality of life.
 In diverse areas such as weather patterns, functioning of brain and operation of a computer.
 In chemical industries

Matter: Anything which occupies space

Classification
 Mixtures: Two or more substances present in any ratio. (air, tea)

Pure Substance: Fixed composition (Cu, Ag)

(i). **Solid:** Particles are held very close to each other in an orderly fashion with no freedom of movement.

Have definite volume and shape.

(ii). **Liquids:** Particles are close and can move around.

Have definite volume but no definite shape.

(iii). **Gases:** Particles are far apart and their movement is easy and fast.

Neither have definite volume nor definite shape.

Properties of Matter:

- (i). **Physical:** Properties measured/observed without changing the identity or composition of substance. (Colour, odour)
- (ii). **Chemical:** Properties measured/observed when a chemical reaction occurs. (Acidity or Basicity, combustibility)

Measurement of Matter:

- International System of Units (SI): (Length- m)
- Mass is the amount of matter present in a substance.
- Weight is the force exerted by gravity on an object.
- Uncertainty of measurement: Range of possible values within which the true value of the measurement lies.
- Significant figures: Meaningful digits known with certainty.
- Precision: Closeness of various measurements for same quantity.
- Accuracy: Agreement of a particular value to true value of result.

Law of Conservation of Mass: Matter can neither be created nor destroyed. It was given by Antoine Lavoisier.

Law of Definite Proportions: A given compound always contains exactly the same proportion of elements. It was given by Joseph Proust.

Law of Multiple Proportions: If two elements can combine to form more than one compound, the masses of one element that combine with a fixed mass of the other element are in ratio of small whole numbers. It was given by Dalton.

Fundamental Equation was developed by Schrodinger as $\hat{H} \psi = E \psi$ where $\hat{H}$ = Hamiltonian

- (i) Principal quantum number (n) : $n = 1, 2, 3, 4, \dots$ Shell = K, L, M, N,
- (ii) Azimuthal Quantum number (l) : For given value of n, $l = 0$ to $n-1$
- (iii) Magnetic Quantum number (m) : for subshell with 'l' value $m_l = 2l+1$
- (iv) Spin Quantum Number (m_s) : $+1/2$ ($\uparrow$), $-1/2$ ($\downarrow$)

- Aufbau Principle: In the ground state of atoms, the orbital's are filled according to increasing energies.
- Pauli Exclusion Principle: No 2 electrons in an atom can have same set of four quantum numbers.
- Hund's rule: Pairing of electrons in the orbitals belonging to same subshell does not take place until each orbital belonging to the subshell is singly occupied.

- Dual behaviour of atom i.e., particle and wavelike. De Broglie equation: $\lambda = \frac{h}{mv} = \frac{h}{p}$
- Heisenberg's Uncertainty Principle: It is impossible to determine simultaneously, the exact position and momentum of an electron

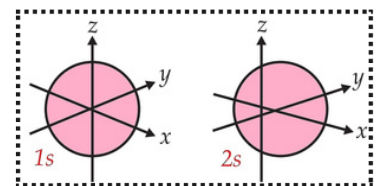
Schrodinger Equation

Quantum Numbers

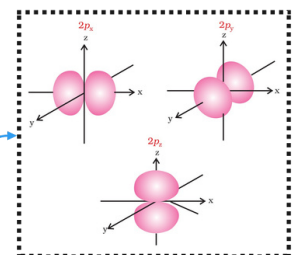
Quantum Mechanical Model of Atom

Filling of orbitals in atom KE of ejected electron $h\nu = h\nu_0 + \frac{1}{2} m v^2$

Shapes of orbitals

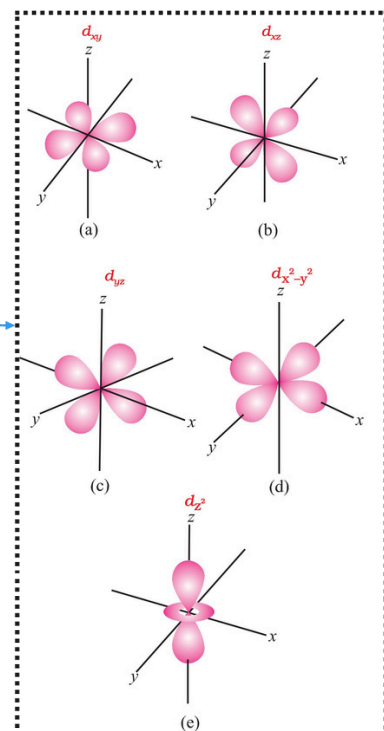


s Orbitals: Spherically symmetric Size increases with increase in size of n, i.e., $4s > 3s > 2s > 1s$



p Orbitals: Each p Orbital consists of two sections called lobes on either side of plane passing through the nucleus.

d Orbitals - "Clover leaf" distribution. - Two angular nodes.



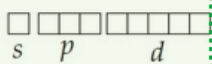
Energy of orbit

- It is the phenomenon of ejection of electrons from the surface of a metal when light of suitable frequency strikes on it.

Lower the value of (n+l) for an orbital, lower is its energy.

Electric Configuration of atoms

(i) $s^a p^b d^c$ notation
(ii) Orbital diagram



Developments leading to Bohr's Model of Atom

- Dual nature i.e. wave like and particle like of the electromagnetic radiation.
- Experimental results regarding atomic spectra.
- Wave nature of electromagnetic radiation: It was given by James Maxwell. Frequency (ν) : Number of waves that pass a given point in one second. Unit - Hertz (Hz), Velocity of light = Frequency $\times$ Wave length
- Wave number (ν) : Number of wavelengths per unit length. Unit - m^{-1}
- Particle nature of electromagnetic radiation: Planck's quantum theory: $E = h\nu$

Planck's constant (h) = 6.626×10^{-34} Js

- Emission Spectra: Spectrum of radiation emitted by a substance that has absorbed energy.
- Absorption Spectra: It is like photographic negative of an emission spectra.
- Line / Atomic Spectra: Emission Spectra which do not show a continuous spread of wavelength from red to violet, rather they emit light only at specific wavelength with dark space between them.

$\bar{\nu} = 109677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ cm}^{-1}$ where
 $n_1 = 1, 2, \dots, n_2 = n_1 + 1, n_1 + 2, \dots$

Series	n_1	n_2	Spectral Region
Lyman	1	2, 3, ...	Ultraviolet
Balmer	2	3, 4, ...	Visible
Paschen	3	4, 5, ...	Infrared
Brackett	4	5, 6, ...	Infrared
Pfund	5	6, 7, ...	Infrared

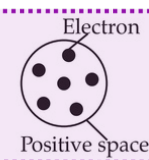
Atoms of different elements with different atomic number but same mass number. (Ca and Ar)

By J.J. Thomson Charge to mass ratio of electron = $1.758820 \times 10^{11} \text{ C kg}^{-1}$
 Charge on electron = $1.6022 \times 10^{-19} \text{ C}$
 Mass of electron = $9.1094 \times 10^{-31} \text{ Kg}$

By Ernest Rutherford
 Charge on proton = $+1.6022 \times 10^{-19} \text{ C}$
 Mass of Proton = $1.672 \times 10^{-27} \text{ Kg}$

By James Rutherford
 Charge on neutron = 0
 Mass of neutron = $1.675 \times 10^{-27} \text{ Kg}$

Atom possesses a spherical shape in which the positive charge is uniformly distributed.



Thomson Model of Atom: (Watermelon Model)

Rutherford's Nuclear Model of Atom

Postulates:

- Positive charge and most of the mass of atoms was densely concentrated in extremely small region (ie) nucleus.
- Nucleus is surrounded by electrons that move around the nucleus with high speed in circular path called orbits.
- Electrons and nucleus are held together by electrostatic forces of attraction.

Drawbacks:

- It cannot explain the stability of an atom.
- It does not say anything about the electronic structure of atoms.

Isobars:

Discovery of Electron

Discovery of Proton

Discovery of Neutron

Structure of Atom

Mass Number (A)

Number of protons (Z) + number of neutrons (n)

Atomic Number (Z)

Number of protons in nucleus of an atom or Number of electrons in a neutral atom

Bohr's Model for Hydrogen Atom

Postulates:

- Electron in H atom can move around the nucleus in a circular path of fixed radius and energy called as orbits. These orbits are arranged concentrically around the nucleus.
- Each of these orbits has a definite energy known as energy levels or stationary states.
- When an electron jumps from a lower energy level to higher one, some energy is absorbed.

$\nu = \frac{E}{h} = \frac{E_2 - E_1}{h}$ Bohr's frequency rule.

Angular momentum of electron: $m_e v r = n \frac{h}{2\pi}$ $n = 1, 2, 3, \dots$

Limitations:

- Unable to account for finer details of H atom. Spectrum observed by sophisticated spectroscopic techniques.
- Could not explain the ability of atoms to form molecules by chemical bonds.

Types:

Atomic Spectra

Isotopes

Atoms of same element having same atomic number but different mass number. (Isotopes of hydrogen)
 Protium $^1\text{H}_1$, Deuterium $^2\text{D}_1$ and Tritium $^3\text{T}_1$

Properties	Group	Period
(a) Atomic Radius : Distance from the centre of the nucleus to the outermost shell containing electrons.	Increases	Decreases
(b) Electron Gain Enthalpy : Energy released when a neutral isolated gaseous atom accepts an electron from anion.	Becomes less negative	Becomes more negative
(c) Ionization Energy: The minimum amount of energy required to remove the electron from the outmost orbit of an isolated atom in gaseous state.	Decreases	Increases
(d) Electronegativity : Tendency of an atom to attract the shared pair of electrons towards it "Group"self.	Decreases	Increases
(e) Electron Affinity	Decreases	Increases
(f) Valency: Number of univalent atoms which combine with an atom of given element.	No Change	Increase from 1 to 4 and then decrease from 4 to 0.
(g) Metallic Character:	Increases	Decreases
(h) Non-Metallic Character:	Decreases	Increases

To ease out difficulty in studying individually the chemistry of all the elements and their compounds.

• Law of Triads: Johann Dobereiner (1829)
 • Law of Octaves: John Alexander Newlands (1865)
 • Periodic Law: Dimitri Mendeleev and Lothar Meyer.
 It states that the properties of the elements are periodic function of their atomic weights.
 • Modern Periodic Law: Henry Moseley (1913)
 It states that the physical and chemical properties of the elements are periodic functions of their atomic numbers. ; Horizontal rows – Periods, Vertical columns – Groups – are numbered from 1 to 18.

Purpose

Genesis of Periodic Classification

IUPAC Nomenclature of Elements with Atomic numbers > 100:

Derived from the atomic number of element using numerical roots for 0 and numbers 1-9 and "ium" is added at the end.

Classification of Elements and Periodicity in Properties

Periodic Trends in Properties of Elements

Properties of Elements based on Electronic Configuration

Periodic Table Classification based on Types of Elements

f- Block: Elements

d- Block: Elements

p- Block: Elements

s- Block: Elements

• Also called as Inner Transition Elements.
 • Contains Lanthanoids and Actinoids.
 • Outer configuration is $(n-2) f^{1-14} (n-1) d^{1-10} ns^{0-2}$
 • All are metals.
 • Actinoids are radioactive.

• Group 3-12
 • Outer configuration is $(n-1) d^{1-10} ns^2 np^6$
 • Forms coloured ions.
 • Exhibit variable valence, paramagnetism.
 • Also called as Transition elements.
 • Some are used as catalysts.

• Group 13 to 18.
 • Also called as representatives or main group elements.
 • Outermost configuration varies from $ns^2 np^1$ to $ns^2 np^6$
 • At the end of period are low reactive noble gases.
 • Halogens and Chalcogens have high negative electron gain enthalpies.
 • Metallic character increases down the group.

• Group 1 (alkali metals) and Group 2 (alkaline earth metals)
 • Outermost configuration is ns^1 or ns^2
 • Reactive with low IE.
 • Metallic character and reactivity increases down the group.

• Electronic Configuration is the distribution of electrons into orbitals of an atom.
 • In periods: Number of elements in each period is twice the number of atomic orbitals available in the energy level that is being filled.
 • Group wise: Elements in same group have similar valence shell electronic configurations. Same number of electrons in outer orbitals and similar properties. These are classified into four blocks i.e., s- block, p- block, d- block and f- block.

Types:

- (i). **Covalent Bond:** A chemical bond formed between two atoms by mutual sharing of electrons between them to complete their octet.
(ii). **Ionic Bond:** A chemical bond formed by complete transference of electrons from one atom to another acquire the stable nearest noble gas configuration.

Energy required to completely separate one mole of a solid ionic compound into gaseous constituent ions.

Postulates:

- Shape of molecule depends upon the number of valence shell electron pairs around central atom.
- Pairs of electrons in the valence shell repel one another.
- These pairs of electrons tend to occupy such positions in space that minimize repulsion.
- The valence shell is taken as a sphere with electron pairs localising on spherical surface at maximum distance from one another.
- A multiple bond is treated as if it is a single electron pair and the two or three electron pairs of a multiple bond are treated as a single super pair.
- When one or more resonance structures can represent a molecule, VSEPR model is applicable:
- **Decreasing order of repulsive interaction:**
 $lp-lp > lp-bp > bp-bp$

Valence Bond Theory: Given by L Pauling. It explains that a covalent bond is formed between two atoms by overlap of their half-filled valence orbitals, each of which contains one unpaired electron.

Orbital Overlap Concept: Formation of a covalent bond results by pairing of electrons in valence shell with opposite spins.

Types of Overlapping: (i) Sigma end. σ bond – end to end. (ii) Pi (TT) bond – axis remain parallel to each other.

Hybridisation: Process of intermixing of orbitals of different energies resulting in formation of new set of orbitals of equivalent energies and shape.

Types of Hybridisation:

Bonding Molecular Orbitals: Addition of atomic orbitals. **Antibonding Molecular Orbitals:** Subtraction of atomic orbitals.

Chemical Bond: Attractive forces which hold the various chemical constituents together in different chemical species.

Lattice Enthalpy

Molecular orbital theory

Chemical Bonding and Molecular Structure

Valence Shell Electron Pair Repulsion (VSEPR)

Chemical Bonding

Hybridization

Bond Parameters

(i) **Bond Length:** Equilibrium distance between the nuclei of two bonded atoms in molecule.

(ii) **Bond Angle:** Angle between the orbitals containing bonding electron pairs around central atom in a molecule complex ion.

(iii) **Bond Enthalpy:** Amount of energy required to break one mole of bonds of particular type between 2 atoms.

(iv) **Bond Order:** Number of bonds between the two atoms of a molecule

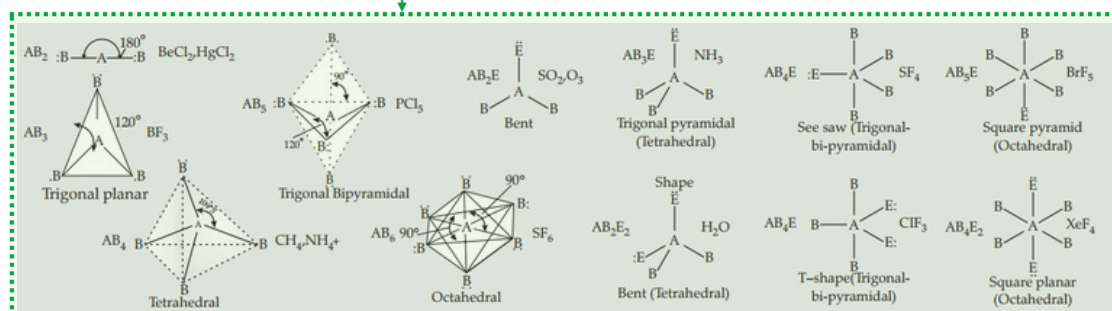
(v) **Resonance Structures:** Are a set of two or more Lewis structures that collectively describe the electronic bonding a single polyatomic species.

(vi) **Dipole Moment:** Product of the magnitude of the charge and distance between centres of positive and negative

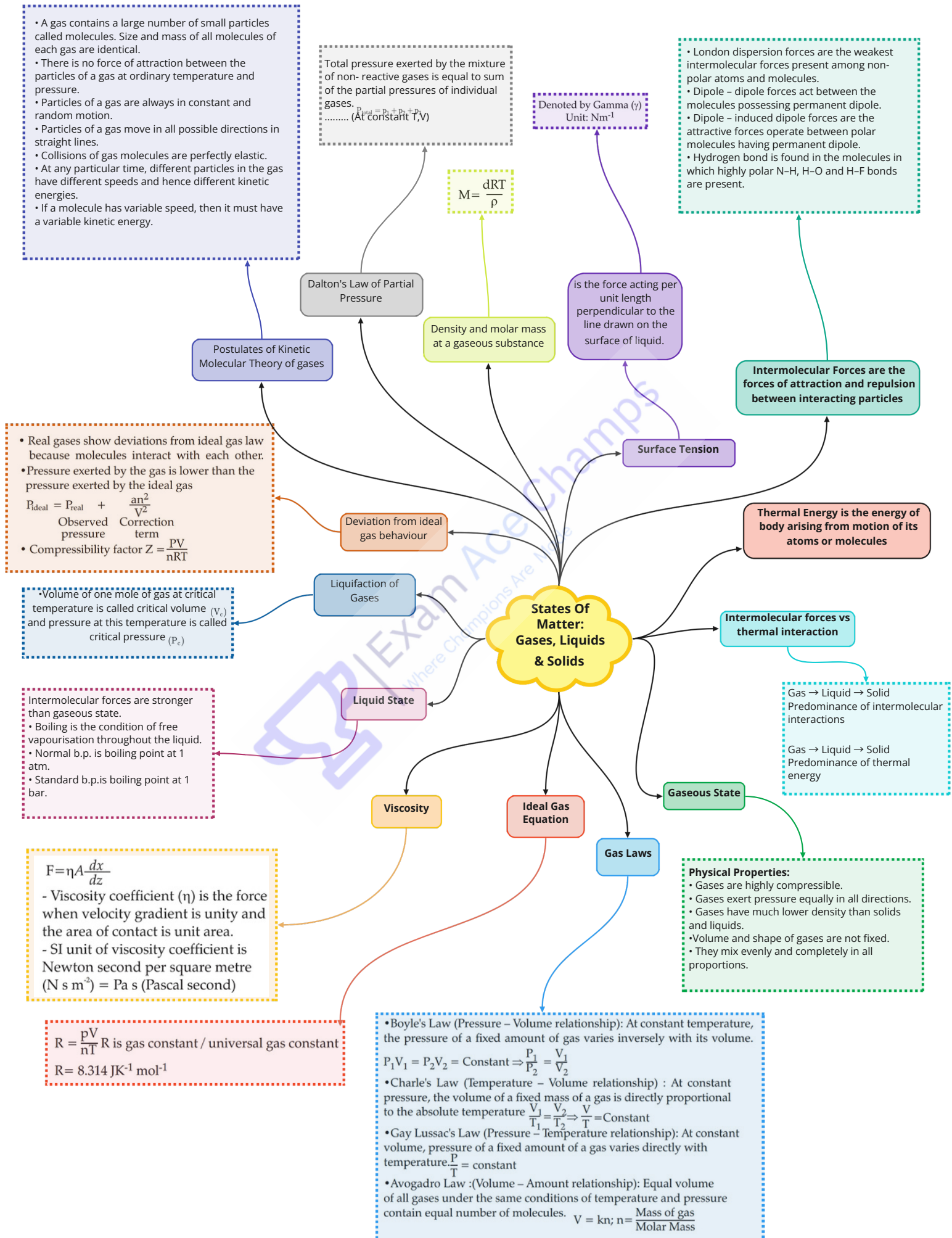
Postulates:

- Electrons in a molecule are present in various molecular orbitals as electrons are present in atomic orbitals.
 - Atomic orbitals of comparable energies and proper symmetry combine.
 - Atomic orbitals is monocentric while a molecular orbital is polycentric.
 - Number of molecular orbital formed is equal to number of combining molecular orbitals.
 - Bonding molecular orbitals has low energy and high stability
- Types of MO: σ (Sigma), π (Pi), δ (Delta)

Kossel Lewis approach to chemical bonding: Lewis pictured the atom as a positively charged 'kernel' and the outer shell accommodates a maximum of eight electrons. • Lewis postulated that atoms achieve the stable octet when linked by chemical bonds. • Kossel gave following facts: In the periodic table, highly electronegative halogens and highly electropositive alkali separated by noble gases. Formation of a negative ion from a halogen atom and a positive ion from an alkali metal atom is associated with gain and loss of electron by respective atoms. Negative and positive ions formed attain noble gas electronic configurations. • Negative and positive ions are stabilized by electrostatic attraction. **Octet Rule:** Atoms can combine either by transfer of valence electrons from one atom to another or by sharing of valence electrons to complete octet in their valence shells. Lewis Dot Structure provides a picture of bonding in molecules and ions in terms of the shared pairs of electrons and the octet rule. **How To Write A Lewis Dot Structure:** Step 1: Add the valence electrons of the combining atoms to obtain total number of electrons. Step 2: For anions, each negative charge means addition of one electron. For cations, each positive charge means subtraction of one electron from total number of valence electrons. Step 3: Write chemical symbols of combining atoms. Step 4: Least electronegative atom occupies central position. Step 5: After accounting for shared pairs of electrons remaining are either utilized for multiple bonding or remain as lone pairs. Formal Charge = (Total number of valence electrons in free atom) – (Total number of non-bonding electrons) – 1/2(Total number of bonding electrons) **Limitations Of Octet Rule:** • Shows three types of exceptions (ie) incomplete octet of central atom, odd-electron molecules and expanded octet. • Does not account for the shape of molecules. • Fails to explain stability of molecules. **Hydrogen Bond:** Formed when the negative end of one molecule attracts the positive end of other. **Types:** (i) Intermolecular: Between two different molecules of same or different compounds. (ii) Intramolecular: H atom is between two highly electronegative atoms.



Class : 11th Chemistry
Chapter- 5: States Of Matter: Gases, Liquids & Solids



$$\Delta_r H = \sum a_i H_{\text{products}} - \sum b_i H_{\text{reactants}}$$

(a) Standard Enthalpy of reactions is the enthalpy change for a reaction when all the participating substances are in their standard states.

(b) Enthalpy changes during phase transformations: Standard enthalpy of fusion / molar enthalpy of fusion, $\Delta_{\text{fus}} H^\ominus$ is the enthalpy change that accompanies melting of one mole of a solid substance in standard state. Standard enthalpy of vaporization or molar enthalpy of vaporization, $\Delta_{\text{vap}} H^\ominus$ is the amount of heat required to vaporize one mole of a liquid at constant temperature and under standard pressure.

(c) Standard molar enthalpy of formation $\Delta_f H^\ominus$ is the standard enthalpy change for the formation of one mole of a compound from its elements in their most stable state of aggregation.

Enthalpy Change $\Delta_r H$ - Reaction Enthalpy

Enthalpy of Solutions
is the enthalpy change
when one mole of a
substance dissolves in a
specified amount of solvent.

Enthalpy of atomization

$\Delta_a H^\ominus$ is the enthalpy change on breaking one mole of bonds completely to obtain atoms in gas phase

Heat Capacity

Applications

Entropy

Gibb's Energy

* $\Delta G = \Delta H - T\Delta S$
 ΔG is < 0 , process is spontaneous.
 * ΔG is > 0 , process is non-spontaneous

$\Delta \mathbf{r} \mathbf{H}^\circ$

$$\Delta H_{\text{rxn}} = \sum \text{bond enthalpies reactants} - \sum \text{bond enthalpies products}$$

$$\rightarrow \Delta_r G_\theta = -RT \ln k$$

Enthalpy, H
 ΔH is negative: Exothermic reactions. ΔH is positive: Endothermic reactions

A spontaneous process is an irreversible process which may be reversed by some external agency

Relationship between C_p and C_v For An Ideal Gas

$$C_p - C_v = R$$

Extensive Property
property which does not depend on the quantity or size of matter present in the system. Temperature, density, pressure.

Reversible Process: The process which can be reversed at any moment by an infinitesimal change.

Irreversible Process: Processes other than reversible process.

At constant temperature $W_{\text{rev}} = 2.303 \text{ nRT} \log \frac{V_t}{V_i}$
For adiabatic change, $q=0$, $\Delta U = W_{\text{ad}}$

- **Systems and surroundings:** A system refers to that part of universe in which observations are made and remaining universe constitutes the surroundings.

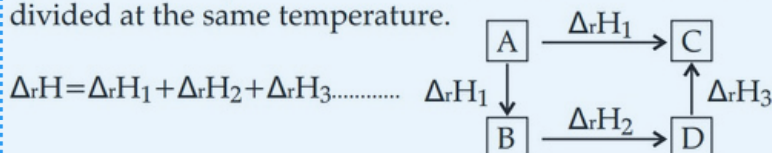
- Types of System:

1. Open System: There is exchange of energy and matter between system and surroundings.
2. Closed System: There is no exchange of matter but exchange of energy is possible.
3. Isolated System: There is no exchange of energy or matter between the system and surroundings.
4. State of system : is described by its measurable or macroscopic properties.
5. Internal Energy: as a state function: It is the sum of chemical, electrical, mechanical or any other type of energy.
6. (a) Work: Adiabatic process is in which there is no transfer of heat between system and surroundings.

- (b) Heat is change in internal energy of a system by transfer of heat from the surroundings to the system or vice-versa without expenditure of work. q is positive: Heat is transferred from surroundings to system. q is negative: Heat is transferred from system to surroundings.
- (c) General Case First law of Thermodynamics: $\Delta U = q + w$ the energy of an isolated system is constant.

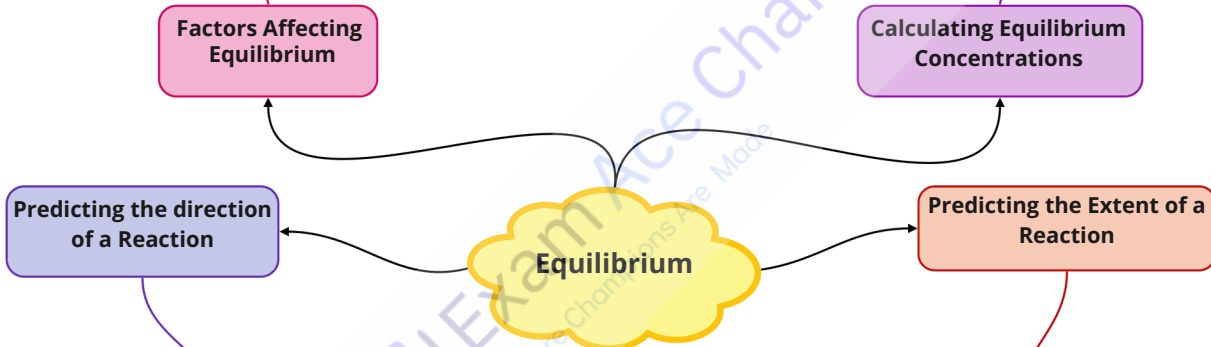
Hess's Law of Constant Heat summation

If a reaction takes place in several steps then its standard reaction enthalpy is the sum of standard enthalpies of international reactions into which the overall reactions may be divided at the same temperature.



• **Le Chatelier's principle:** A change in any of the factors that determine the equilibrium conditions of a system will cause the system to change in such a manner so as to counteract the effect of the change. • **Effect of Concentration Change:** If concentration of reactants is increased, equilibrium will shift in forward direction and vice- versa. • **Effect of Pressure Change:** No effect if moles of reactants and products are same. If there is change in number of moles, equilibrium will shift in the direction having smaller number of moles when the pressure is increased and vice- versa. • **Effect of Inert Gas Addition:** If volume is constant, i.e., addition does not change partial pressure or molar concentrations, equilibrium is not affected. • **Effect of Temperature change:** For exothermic, low temperature favours formation of reactants; for endothermic, high temperature favours formation of products.

Step 1: Write the balanced equation for the reaction.
Step 2: Make a table that lists for each substance involved in reaction; Initial concentration, change in concentration, equilibrium concentration.
Step 3: Substitute the equilibrium concentrations into the equilibrium equation for the reaction and solve.
Step 4: Calculate equilibrium concentrations from the calculated value of concentration of substances.
Step 5: Check results by substituting them into the equilibrium equation.
Buffer Solutions are the solutions which resist change in pH on dilution or addition of acid or alkali.
Solubility Product (K_{sp}): For a sparingly soluble salt, it is the product of molar concentration of ions raised to power equal to numbers of times each ion.



- If $Q_c < K_c$, net reaction goes from left to right.
- If $Q_c > K_c$, net reaction goes from right to left.
- If $Q_c = K_c$, no net reaction occurs

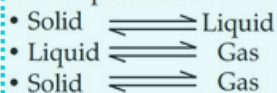
- $K_c < 10^{-3}$ reaction proceeds rarely.
- $K_c > 10^3$ reaction proceeds nearly to completion
- $K_c \rightarrow 10^{-3}$ to 10^3 , reaction is at equilibrium

Reactants and products are in same phase.

Types

Reactants and products are present in two or more phases.

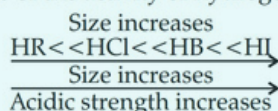
Equilibrium set up in physical processes like evaporation of water, etc.



Characteristics:

- Possible only in a closed system at a given temperature.
- Both the reactions occur at same rate.
- All measurable property remains constant.
- When equilibrium is attained one of its parameter remains constant at given temperature.
- Magnitude of such quantities at any stage indicates the extent to which process has proceeded before reaching equilibrium.

pH of a solution is negative logarithm to base 10 of the activity of hydrogen ion



Speed of forward reaction equals that of backward reaction.

Characteristics:

- Can be achieved from either side.
- Dynamic in nature.
- There is no change in concentration of various species.
- Observable physical properties become constant.
- Equilibrium is attained rapidly by the presence of catalyst.

Step 1: Write the correct formula for each reactant and product. **Step 2:** Identify atoms undergoing change in O.N. **Step 3:** Calculate increase/decrease in O.N. per atom and for entire ion or molecule. If unequal, multiply by suitable number to make equal. **Step 4:** Add

H^+/OH^- ion to make total ionic charges of reactants and product equal.

Step 5: Equalize H^+ on two sides by adding water.

O.N. Method

Half- Reaction Method

Two Methods

Step 1: Produce, unbalanced equation for the reaction in ionic form.

Step 2: Separate equation into two half- reactions.

Step 3: Individually balance atoms other than O and H.

Step 4: For reactions occurring in acidic medium, add H_2O to balance O atoms and H^+ to balance H atoms.

Step 5: Balance charges by adding electrons to one side of the half reaction.

Step 6: Add two half reactions and cancel the electrons on each side.

Step 7: Verify if equation has same type and number of atoms and same charges on both sides of the equation.

Redox Reactions and Electrode Processes

Study of electrode processes and cells.

Redox Couple: Having together oxidised and reduced forms of a substance involved in oxidation/reduction half reaction.

Electrode Potential: Potential associated with each electrode.

Standard Electrode Potential: Electrode potential of concentration of each species is unity and carried out at 298K.

Negative E: Stronger reducing agent than H^+/H_2 couple.

Positive E: Weaker reducing agent than H^+/H_2 couple.

Rules for O.N.

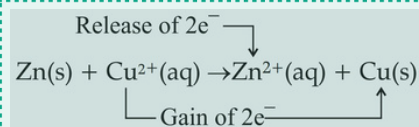
Oxidation state of an element in a compound ascertained according to a set of rules formulated on the basis that pair in a covalent bond belongs to electronegative element.

1. In elements, each atom bears an oxidation number of zero.
2. For ions with only one atom, O.N. is equal to the charge on ion.
3. O.N. of oxygen in most compounds is -2.
4. O.N. of hydrogen is +1, except when it is bonded to metals in binary compound.
5. Halogens have on O.N. of -1, when they occur as halide ions in their compound.
6. Algebraic sum of O.N. of all the atoms in a compound must be zero.

Oxidation Number

Balancing of Redox Reactions

Competitive Electron Transfer Reactions



Redox Reactions

Oxidation: Addition of oxygen to an element/ compound or removal of hydrogen from a substance

Reduction: Removal of oxygen from a substance or addition of hydrogen to a substance

Loss of electrons

Reducing agent: Donor of electron(s)

Lowers O.N.

Gain of electrons

Oxidising agent: Acceptor of electron(s)

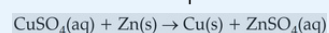
Increases O.N.

Application

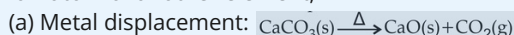
Types of Redox Reactions

1. Combination reactions: $\text{A} + \text{B} \rightarrow \text{C}$; Either A and B or both A and B must be in elemental form.

2. Decomposition reactions: Leads to the breakdown of a compound into two or more components at least one of which must be in the elemental state



3. Displacement reactions: An ion/ atom in a compound, is replaced by an ion/atom of another element, $\text{X} + \text{YZ} \rightarrow \text{XZ} + \text{Y}$



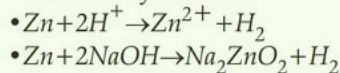
(b) Non- Metal displacement: Includes hydrogen displacement.

4. Disproportionation reaction: An element in one oxidation state is simultaneously oxidised and reduced.

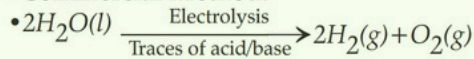
Occurrence: Most abundant in universe.

Preparation:

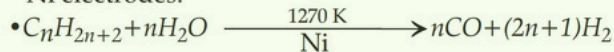
– Laboratory Method:



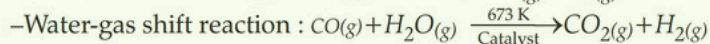
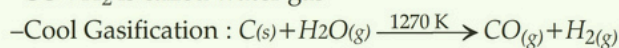
– Commercial Method:



• By electrolysis of warm aqueous barium hydroxide solution between Ni electrodes.



– CO + H₂ is called water gas

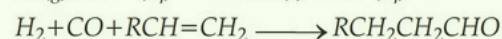
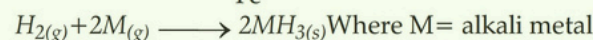
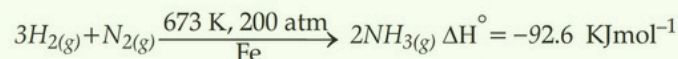
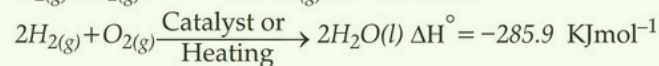
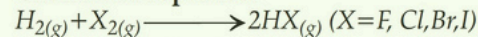


Physical Properties:

• Colourless, odourless, tasteless and combustible.

• Lighter than air, insoluble in water.

Chemical Properties:



Uses:

- Synthesis of ammonia.
- In manufacture of vanaspathi fat.
- In manufacture of bulk organic chemicals.
- For manufacture of metal hydrides
- Preparation of HCl.
- In metallurgical processes.
- As a rocket fuel
- In fuel cells

Preparation: By exhaustive electrolysis of water.

Uses:

- As a moderator in nuclear reactors.
- In exchange reactions for the study of reaction mechanisms.

Heavy Water (D₂O)

Protium: Predominant form. (¹H)
Deuterium: (²H)
Tritium: Radioactive (³H)

Isotopes

- Resembles alkali metals (lose one e⁻ to form unipositive ions).
- Resembles halogens (gain one e⁻ to form uninegative ion).
- Forms oxides, halides and sulphides.
- Very high ionization enthalpy.
- Does not possess metallic characters under normal conditions.
- Forms diatomic molecules

Properties

Types:

- Ionic/saline/saltlike:** Stoichiometric compounds of dihydrogen formed with s- block elements.
- Covalent/molecular:** Formation of molecular compounds from dihydrogen and p- block elements.
- Metallic/non- stoichiometric/interstitial:** Formed by d- block and f- block elements.

Hydrides

Water

Hydrogen

Electronic Configuration
1s¹

Dihydrogen (H₂)

Hydrogen Peroxide (H₂O₂)

Preparation

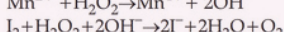
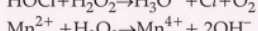
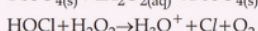
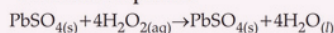
Physical Properties

Uses

• Colourless, miscible with water.

Structure: Non-Planar

Chemical Properties:



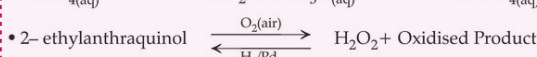
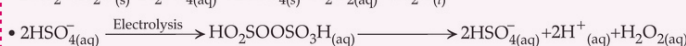
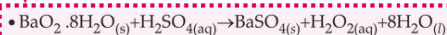
Decomposes on exposure

• As hair bleach, disinfectant.

• Manufacture chemicals used in detergents.

• In industries as bleaching agent.

• In environmental chemistry.

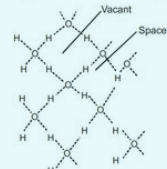
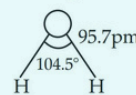


(Human body: 65%, Plants: 95%)

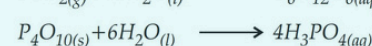
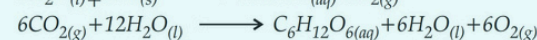
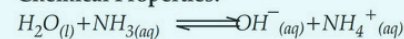
Physical Properties:

- Colourless and tasteless
- Presence of extensive H bonding.
- High freezing point, high B.P, high heat of vaporization, high heat of fusion

Structure:



Chemical Properties:



Difference: Hard water does not give lather with soap whereas soft water gives lather easily. Temporary hardness: Removed by boiling, Clark's method. (Due to presence of magnesium and calcium hydrogen carbonates)

Permanent hardness: Removed by treatment with washing soda, Calgon's method, Ion exchange method, synthetic resins methods. (Due to presence of soluble salts of Mg and Ca in the form of chlorides and sulphate)

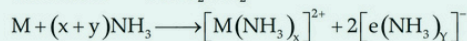
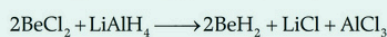
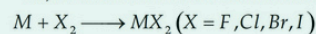
- **Atomic and Ionic Radii:** Smaller than corresponding alkali group, increases with increase in atomic number.
- **I.E:** IE₁ higher than corresponding Group 1 metals. IE₂ smaller than corresponding alkali metals.
- **Hydration Enthalpies:** Decreases with increase in ionic size down the group.

Physical properties:

- Silvery white, lustrous and relatively soft but harder than alkali metals.
- M.p. and b.p. higher than corresponding alkali metals.
- Electropositive character increases down the group.

Chemical properties:

- Be and Mg are kinetically inert to O and H₂O
- Mg is more electropositive and burns in air.
- Ca, Sr and Ba with air forms oxide and nitride.



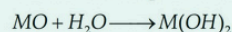
Uses:

- Be is used in the manufacture of alloys.
- Metallic Be is used for making windows of X-rays tubes.
- Mg-Al alloys are used in air craft construction.
- Ca in extraction of metals.
- Ra is used in radiotherapy.

Characteristics of Compounds of Alkaline Earth Metals:

• **Oxides and Hydroxides**

- Alkaline earth metals burn oxygen to form MO.
- All oxides except BeO are basic in nature



Be(OH)₂ is amphoteric in nature

Halides:

- Except for Be halides, all other halides are ionic.
- Tendency to form halide hydrates decreases gradually.
- Salts of oxoacids: Forms carbonates, sulphates and nitrates.
- Anomalous behavior of Be: Small atomic and ionic sizes, does not exhibit C.N. more than four, its oxide and hydroxide are amphoteric.
- Be shows diagonal relationship with Al.

Biological Importance Of Mg And Ca:

- All enzymes that utilise ATP in PO₄ transfer requires Mg as cofactor. Chlorophyll contains Mg. Ca is present in bones and teeth. Important in neuromuscular function, intraneuronal transmission and blood coagulation.

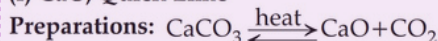
Group 2 Elements (Alkaline Earth Metals)

Group 1 Elements (Alkali Metals)

S- Block Elements

Important Compounds of Calcium

(i) CaO, Quick Lime

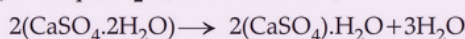


Properties: White amorphous solid with m.p. 2870 K
 $CaO + H_2O \longrightarrow Ca(OH)_2$, $CaO + CO_2 \longrightarrow CaCO_3$

(ii) Ca(OH)₂ Calcium hydroxide: Preparations: Additional of water to CaO.

Properties: white amorphous powder.
 $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$

(iii) CaSO₄ · ½H₂O (Plaster of Paris)



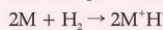
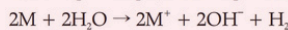
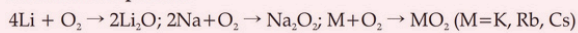
Electronic Configuration

- **Atomic and Ionic Radii:** Increases with increase in atomic number.
- **I.E.:** Decreases down the group.
- **Hydration Enthalpy:** Decreases with increase in ionic sizes.

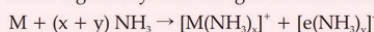
• **Physical properties:**

- Silvery white, soft and light metals.
- Low m.p. and b.p.
- Alkali metals and their salts impart colour to an oxidizing flame.

Chemical Properties:



React vigorously with halogens to form ionic halides

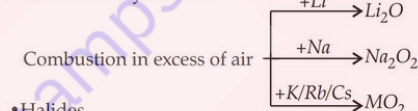


Uses:

- Li is used to make useful alloys.
- Li is used in thermonuclear reactions and making electrochemical cells.
- Na is used to make Na/Pb alloy.
- Liquid Na metal is used as coolant in nuclear reactors.
- KCl is used as fertilizer.
- Cs is used in devising photoelectric cells.

Characteristics Of Compounds Of Alkali Metals:

– Oxide and Hydroxides



• **Halides**

- Alkali metal halides (MX) have high melting, colourless crystalline solids.
- Preparation: Reaction of oxide, hydroxide or carbonate with aq HX.
- High negative enthalpies of formation.
- Melting and boiling points: F > Cl > Br > I
- Soluble in water.

Salts of Oxo-Acids:

- Alkali metals form salts with all oxo-acids.
- Soluble in water and thermally stable.
- Stability of carbonates and hydrogencarbonates increases

Anomalies properties of Li: Due to

- exceptionally small size of its atom and ion.
- High polarising power.

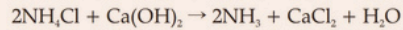
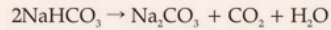
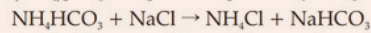
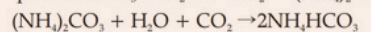
Biological Importance of Na and K:

Na ions participate in nuclear signals transmission, regulator of flow of water across cell membranes. K ions activate many enzymes and oxidation of glucose to produce ATP.

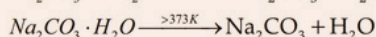
ns¹ : Alkaline metals; ns² : Alkaline earth metals

Important Compounds of Sodium:

(i) **Sodium Carbonate (Washing Soda):** Preparation: By Solvay process
 $2NH_3 + H_2O + CO_2 \rightarrow (NH_4)_2CO_3$



Properties: (a) White, crystalline solid. (b) Readily soluble in water
 $Na_2CO_3 \cdot 10H_2O \xrightarrow{375K} Na_2CO_3 \cdot xH_2O + 9H_2O$



Uses: Water softening, laundering, cleaning, manufacture, as laboratory reagent.

(i) **Sodium Chloride (NaCl)**

Preparation: Crude NaCl by crystallization of brine solution.

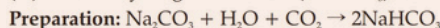
– Pure NaCl is obtained by dissolving crude salt in minimum water and filtered to remove insoluble impurities. Solution is saturated with HCl gas. **Uses:** As common salt.

(ii) **Sodium Hydroxide (NaOH)**

Preparation: By electrolysis of NaCl in Castner-Kellner cell.

Uses: In manufacture of soap, paper, petroleum refining.

(iii) **Sodium Hydrogencarbonate (NaHCO₃)**



[B, Al, Ga, In, Tl]

Electronic configuration: $ns^2 np^1$

Atomic Radii: Atomic radius of Ga is less than Al.

I.E.: Decrease from B to Al is associated with increase in size. $\Delta H_1 < \Delta H_2 < \Delta H_3$

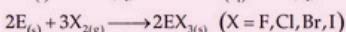
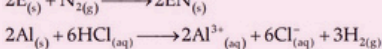
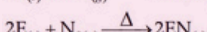
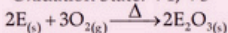
Electronegativity: Down the group, decreases from B to Al and then increases marginally.

Physical Properties:

- Non-metallic, hard and black coloured solid.
- Exists in many allotropic forms.
- B has high m.p. while others have low m.p.
- Density increases down the group.

Chemical Properties:

• Oxidation State: +1, +3

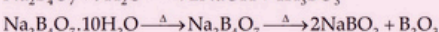
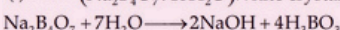


Trends and Anomalous Properties of Boron:

- Except in boron, $[M(OH)_4]^-$ and $[M(H_2O)_6]^{3+}$ exist in aqueous medium.
- Boron trifluoride reacts with Lewis bases.
- Maximum covalence of B is 4.

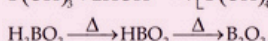
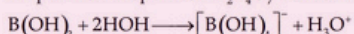
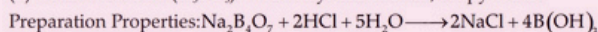
Important Compounds of Boron:

(i) Borax ($Na_2B_4O_7 \cdot 10H_2O$) White crystalline solid

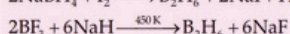
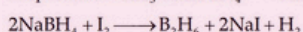
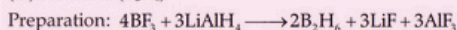


Borax bead test is used for identification.

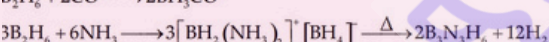
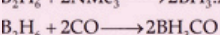
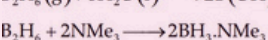
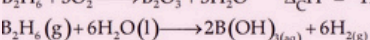
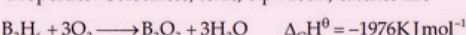
(ii) Orthoboric acid (H_3BO_3): White crystalline solid, soapy touch



(iii) Diborane (B_2H_6)



Properties: Colourless, toxic, b.p.: 180K, catches fire



Uses of Boron and Aluminium and their Compounds:

- Boron fibers are used in making bullet-proof vest and light material for aircraft.
- Borax is used in manufacturing heat resistant glasses.
- Al is used in packing, utensil making, construction and transportation industry.

[C, Si, Ge, Sn, Pb]

Electronic configuration: $ns^2 np^2$

Covalent Radius: Increase from C to Si and from Si to Pb a small increase is observed.

I.E.: Decreases down the group.

Electro negativity: More than group 13 elements. From Si to Pb same.

Physical Properties:

- Solid.
- C and Si are non- metals, Ge metalloid, Sn and Pb are soft metals.
- M.p. and b.p. higher than group 13 elements.

Chemical Properties:

- Oxidation State: +4, +2
- Forms oxides on heating in oxygen.
- C, Si, Ge, and Pb are not affected by water.



• Form halides of formula MX_2 and MX_4

Important Trends and Anomalous Behavior of Carbon:

• C differs from rest due to smaller size, higher electronegativity, high ionisation enthalpy and unavailability of d orbitals.

• C forms $p\pi-p\pi$ multiple bonds

• Shows Catenation

Allotropes of Carbon:

- Diamond
- Carbon
- Fullerenes

Uses of Carbon:

- Graphite composites are used in tennis rackets, fishing rods, aircrafts and canoes.
- Graphite is used for electrodes in batteries and industrial electrolysis.
- C black is used in black ink and as filter in automobile tyres.
- Diamond is used in jewelry.

Group 14
Elements

Group 13
Elements

p - Block
Elements

Important Compounds of C and Si

Zeolites: used as catalyst in petrochemical industries.

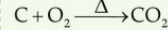
Silicon Dioxide: Si is covalently bonded to 4 oxygen atoms.

Silicates Structural 4- unit is

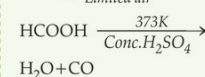
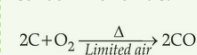
Silicones: Group of organosilicones

polymers $(R_2SiO)_n$ with replanting unit.

Carbon Dioxide:



Carbon monoxide:



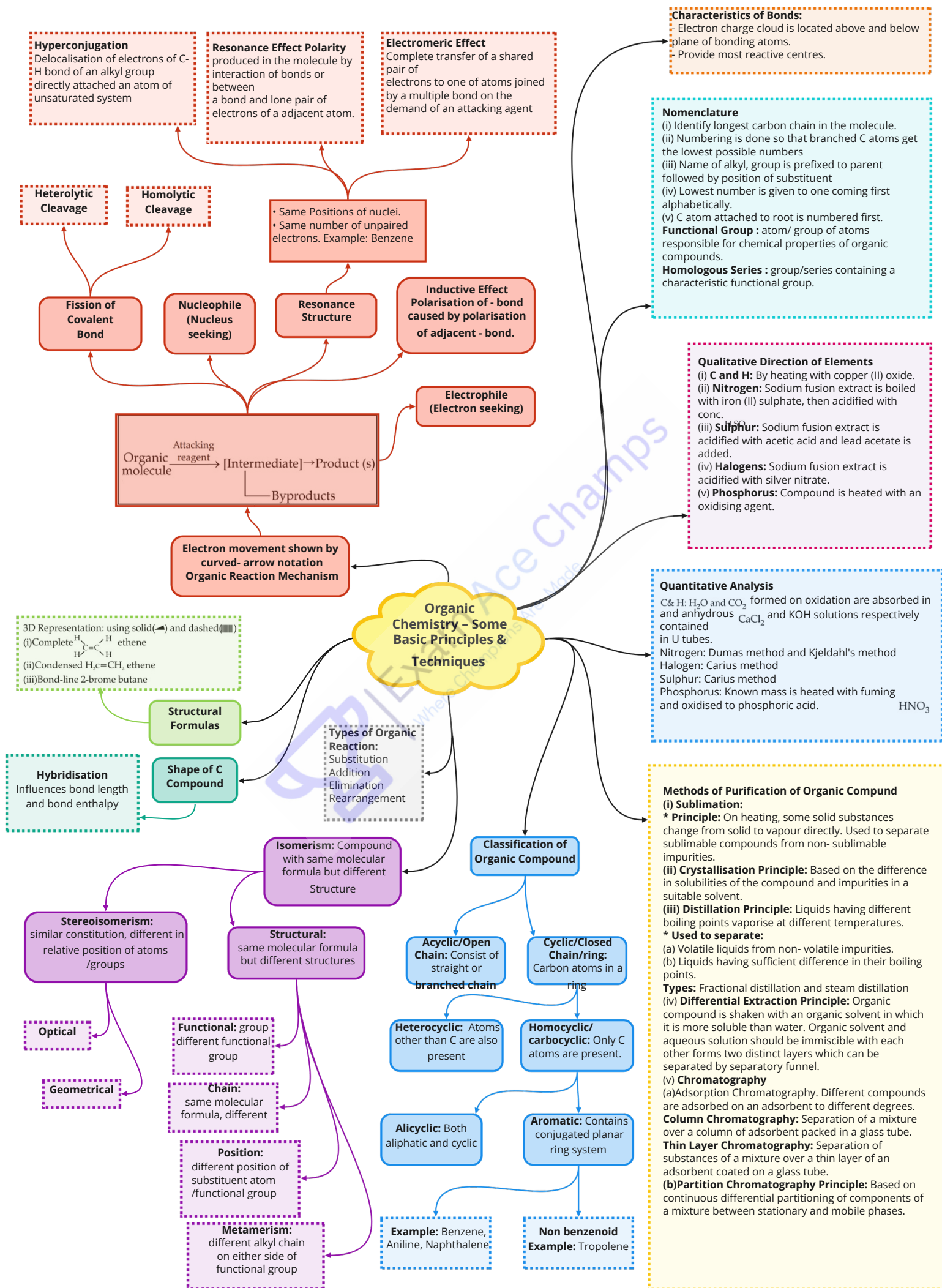
Electronic configuration:

$ns^2 np^{1-6}$ (except He)

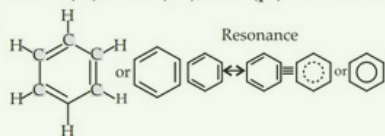
Producer Gas: Mixture of $CO + N_2$

Water Gas: or Synthesis Gas:
Mixture of $CO + H_2$

Class : 11th Chemistry
Chapter - 12: Organic Chemistry – Some Basic Principles & Techniques



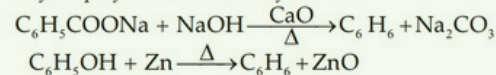
- Types: Benzenoids – contain benzene ring, Non-benzenoids – does not contain benzene ring.
- Isomerism: Ortho (o-), Meta (m-), Para (p-) Resonance and
- Structure:



- Aromaticity: Planarity, complete delocalisation of the π -electrons in the ring, presence of $(4n + 2)$ π electrons in the ring where n is an integer ($n = 0, 1, 2, \dots$) (Huckel rule)

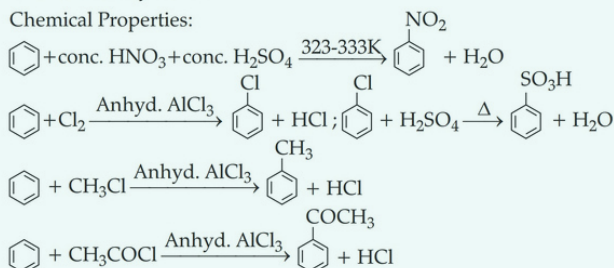
Preparation:

- Cyclic polymerisation of ethyne



- Non-polar, usually colourless liquids or solids with characteristic aroma.
- Immiscible with water but miscible with organic solvents.
- Burns with sooty flame.

Chemical Properties:



Classification

- Saturated:** Contain C-C and C-H single bonds. (alkanes)
- Unsaturated:** Contain C-C multiple bonds.
- Aromatic:** Contain Cyclic Compounds

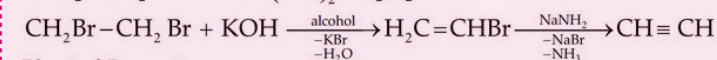
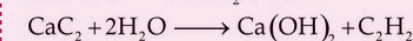
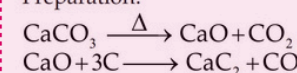
Mechanism of Electrophilic substitution reactions

- Generation of Electrophile
- Formation of carbocation intermediate
- Removal of proton

- IUPAC name: replacing 'ane' by the suffix 'yne'.

- Shows position and chain isomerism

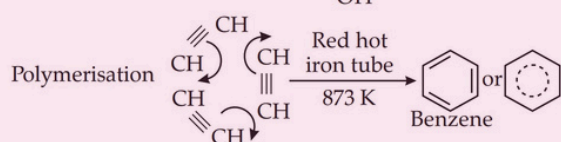
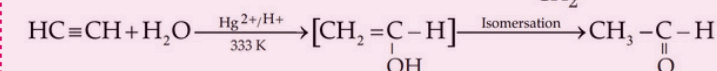
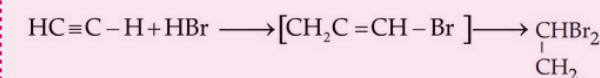
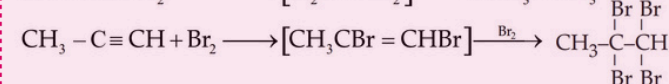
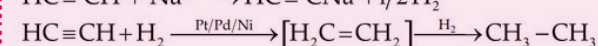
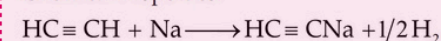
Preparation:



Physical Properties:

- First three members are gases, next eight are liquids and higher ones are solids.
- Colourless, ethyne has characteristic odour and other are odourless.
- Lighter than water, immiscible with water but soluble in organic solvents.
- m.p, b.p. and density increase with increase in molar mass.

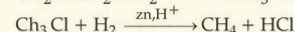
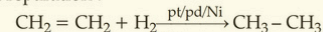
Chemical Properties:



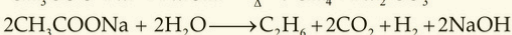
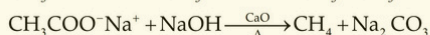
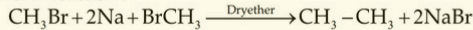
- H-C-H bond angles – 190.5, C-C and C-H bond lengths are 154 pm and 112 pm respectively.

- Shows structural and chain isomerism.

Preparation :



Wurtz reaction :

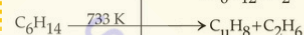
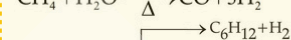
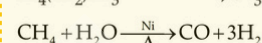
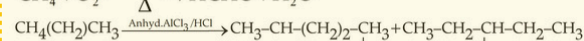
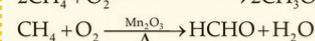
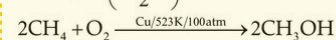
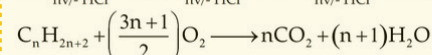
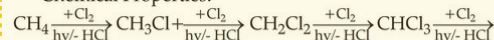


- Physical Properties:

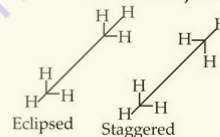
Non-polar, weak van der Waals forces, colourless, odourless.

B.P increases with increases in molecular size.

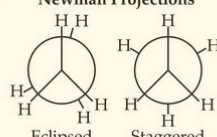
Chemical Properties:



Seahorse Projections



Newman Projections



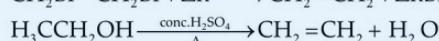
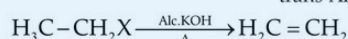
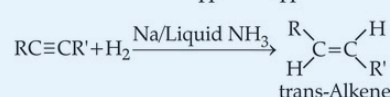
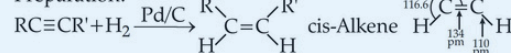
Hydrocarbons

Alkene($\text{C}_2\text{H}_{2n-2}$)

Alkenes(C_2H_{2n})

- Shows structural and geometrical isomerism

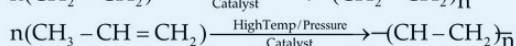
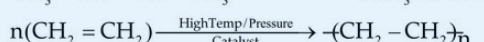
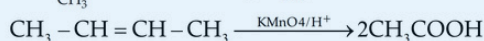
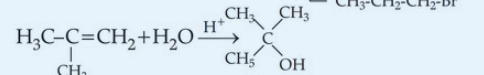
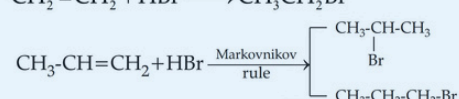
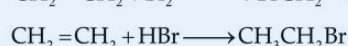
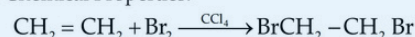
Preparation:

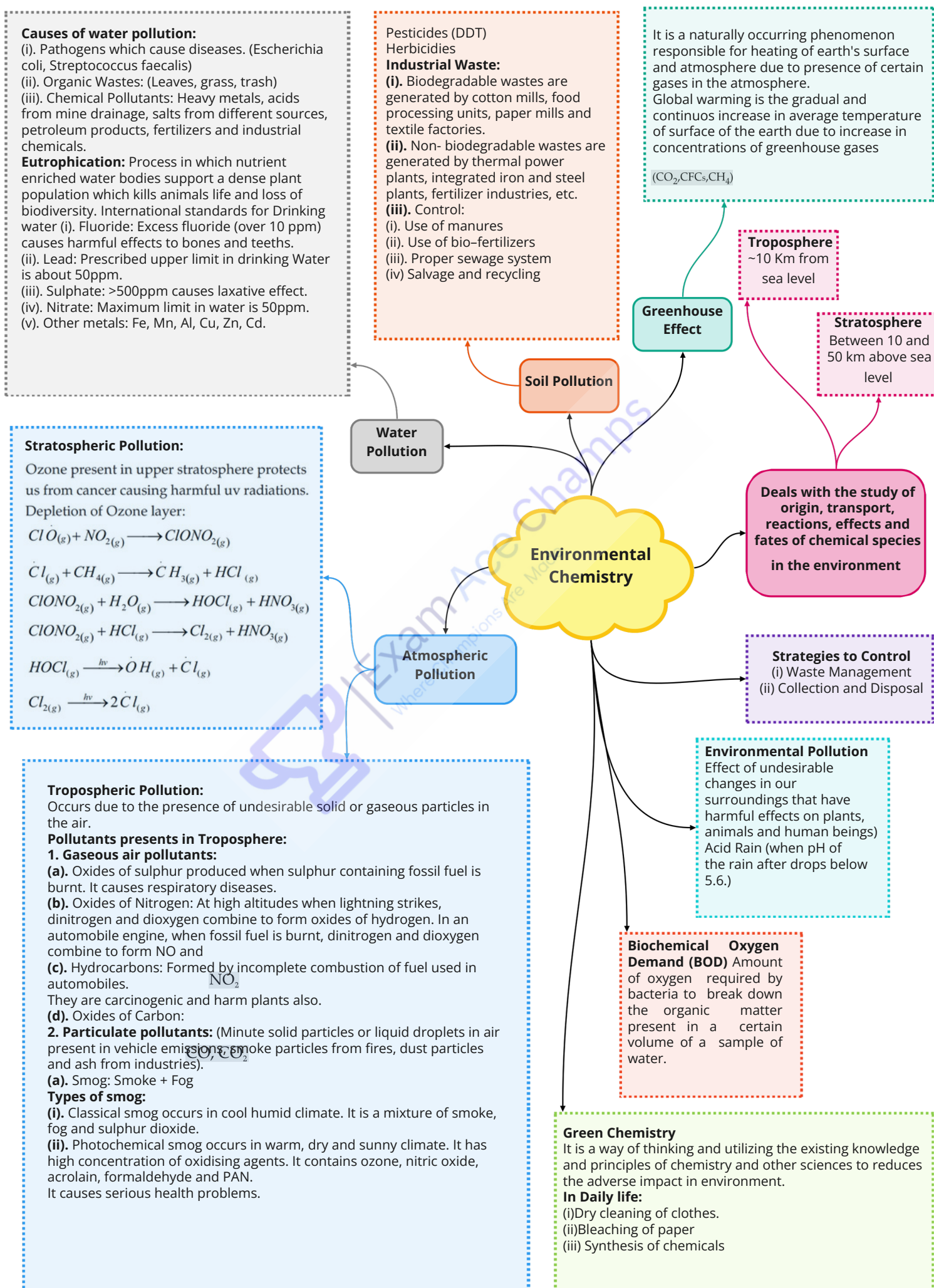


- Physical Properties :

- Ethene is a colourless gas with faint sweet smell.
- All other are colourless and odourless, insoluble in water but fairly soluble in non-polar solvents.
- Increase in b.p. with increased molecular size.

- Chemical Properties:







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